

Synchronous buck converter design

1. **Objective:** Simulation and analysis of an open-loop 60W synchronous buck converter in LTspice incorporating practical parasitic elements and switching transient mitigation techniques for efficiency improvement over an asynchronous buck design.

2. **Specifications:**

- 2.1 Input voltage (V_{in}): 42 V
- 2.2 Output voltage (V_{out}): 12 V
- 2.3 Output Current (I_{out}): 5 A
- 2.4 Output power (P_{out}): 60 W
- 2.5 Switching frequency (f_{sw}): 200 kHz
- 2.6 MOSFET: BSC190N12NS3(Infineon)
- 2.7 Load resistance (R_l): 2.4Ω
- 2.8 Input source resistance: $100m\Omega$
- 2.9 Input parasitic inductance: $3nH$
- 2.10 Inductor DCR: $50m\Omega$
- 2.11 Capacitor ESR: $30m\Omega$
- 2.12 Output capacitor: $22\mu F$
- 2.13 Gate resistance V2 & V3: 10Ω
- 2.14 Snubbers (RC): $10\Omega / 1nF$
- 2.15 High Side (HS) on time: $1.469\mu s$
- 2.16 Low Side (LS) delay: $1.499\mu s$
- 2.17 Low side (LS) on time: $3.471\mu s$
- 2.18 Dead time: 30ns

3. **Design calculation:**

3.1 Baseline power stage reference: The passive power stage component values and switching frequency were retained from previously developed asynchronous buck converter design, as the required electrical specifications remained unchanged. The transition to synchronous rectification primarily involved replacing the freewheeling diode with a low side MOSFET and re-optimizing gate timing to improve efficiency and reduce conduction losses. While the inductor design rationale, CCM boundary verification and passive component selection are documented in the companion asynchronous buck converter report.

3.2 Ideal Duty Cycle and Initial High-Side ON-Time Estimation: For an ideal buck converter operating in continuous conduction mode, the duty cycle is given by:

$$D_{ideal} = \frac{V_{out}}{V_{in}}$$

Substituting the design values: $D_{ideal} = \frac{12}{42} = 0.2857$

The switching period corresponding to 200kHz operation is : $T_s = \frac{1}{f_s} = \frac{1}{200000} = 5\mu s$

Therefore the ideal theoretical high-side MOSFET ON-Time is:

$$T_{ON,ideal} = D_{ideal} * T_s = 0.2857 * 5 = 1.428 \mu s$$

The value served as the initial theoretical reference for high-side gate pulse generation.

3.3 Practical ON-Time optimization under Non-ideal conditions: When simulated with practical parasitic elements and non-ideal component models the theoretical ON-Time of 1.428 μs resulted in output undervoltage due to conduction and switching losses introduced by:

- MOSFET RDS (on)
- Inductor DCR
- Capacitor ESR
- Dead-time body diode conduction
- Input parasitic inductance
- Switching transition losses

To compensate for these practical losses, the high side MOSFET ON-time was increased iteratively through simulation until the output voltage converged to the target value of 12V.

Final optimized value selected:

$$T_{ON,HS} = 1.469\mu s$$

This optimized ON-time produced the desired regulated output under full-load conditions.

3.4 Dead-Time selection Selection and Low-side timing derivation: To prevent simultaneous conduction of the high-side and low-side MOSFETs (shoot-through), a non-overlap dead time of 30ns was introduced between switching transitions.

Selected dead-time: $T_{dead} = 30ns$

The low-side MOSFET On-time delay was calculated as:

$$T_{delay,LS} = T_{ON,HS} + T_{dead} = 1.469 + 0.030 = 1.499\mu s$$

The low-side MOSFET ON-time was determined by subtracting the high-side ON-time and both dead time intervals from the total switching period:

$$T_{ON,LS} = T_s - T_{on,HS} - 2T_{dead} = 5 - 1.469 - 0.060 = 3.471\mu s$$

Thus the final low-side gate timing parameters selected as:

- **Low-side Turn-on delay: 1.499 μ s**
- **Low-side ON-time: 3.471 μ s**

This ensured symmetrical dead-time at both switching transitions

3.5 Switching Transient Mitigation and snubber optimization: Initial synchronous switching simulations exhibited significant voltage overshoot at the switch node due to high di/dt and parasitic input loop inductance.

To mitigate transient overshoot and ringing:

- Gate resistance was optimized to control switching slew rate
- Input parasitic inductance was minimized to represent improved layout conditions
- An RC snubber network was added across switching node.

Final transient mitigation parameters:

- **Gate resistance: 10 Ω**
- **RC Snubber: 10 Ω & 2nF**
- **Input parasitic inductance: 3nH**

These measures reduced switch-node overshoot to an acceptable level while maintaining safe voltage margin below the MOSFET drain-source rating.

4. LTspice circuit implementation:

4.1 Circuit overview: The synchronous buck converter was implemented in LTspice using a two-MOSFET switching topology to replace the conventional freewheeling diode employed in the asynchronous buck converter. The complete simulation model incorporates practical parasitic elements and transient mitigation components to closely emulate real-world converter behaviour.

4.2 High-Side MOSFET Switching Stage: The high-side MOSFET functions as the primary controlled switching device, periodically connecting the 42 V input source to the switching node based on the applied PWM gate pulse. It regulates the energy transferred to the output stage by controlling the effective duty cycle of the converter.

4.3 Low-Side Synchronous MOSFET: The low-side MOSFET replaces the freewheeling diode used in the asynchronous topology and provides a low-resistance current path for inductor freewheeling during the OFF-state of the high-side MOSFET. This significantly reduces conduction losses associated with diode forward voltage drop, thereby improving converter efficiency.

4.4 Output LC Filter: The output LC filter consists of a 33 μ H inductor and 22 μ F output capacitor. The inductor smooths the pulsating switch-node current, while the capacitor filters output voltage ripple to maintain a stable DC output across the load.

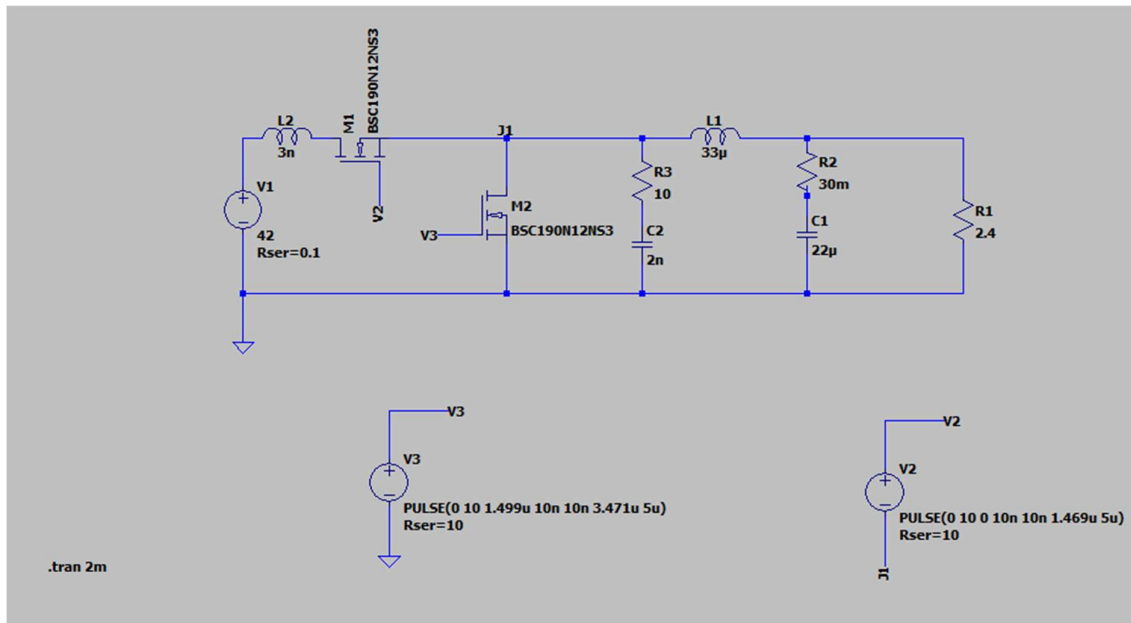
4.5 RC Snubber Network: An RC snubber network was connected across the switching node to suppress high-frequency ringing and voltage overshoot caused by parasitic inductance and rapid switching transitions. The selected snubber values were optimized through iterative simulation to balance transient suppression and power dissipation.

4.6 Included Parasitic Elements: To improve simulation realism, practical parasitic elements were incorporated into the converter model, including:

- Input source resistance to represent source and interconnect losses
- Input parasitic inductance to model PCB trace and loop inductance
- Inductor DCR to account for winding resistance
- Capacitor ESR to model non-ideal capacitor losses
- Gate resistance to emulate practical gate drive impedance
- These parasitic elements significantly affect converter transient response, voltage overshoot, and overall efficiency, and were therefore included in the design model.

4.7 Gate Pulse Generation Sources: Independent pulse voltage sources were used to generate complementary gate drive signals for the high-side and low-side MOSFETs. The gate timing was configured to maintain a controlled dead-time between switching transitions, preventing shoot-through while minimizing body-diode conduction losses.

4.8 Circuit visual:



5. Simulation result and analysis:

5.1 Gate drive timing verification: The gate-drive waveforms of the high-side V(V2(j1)) and low-side MOSFETs V(v3) were analysed to verify proper synchronous switching operation of the converter. Complementary PWM signals were applied to both MOSFETs with a controlled dead-time to prevent simultaneous conduction and shoot-through current.

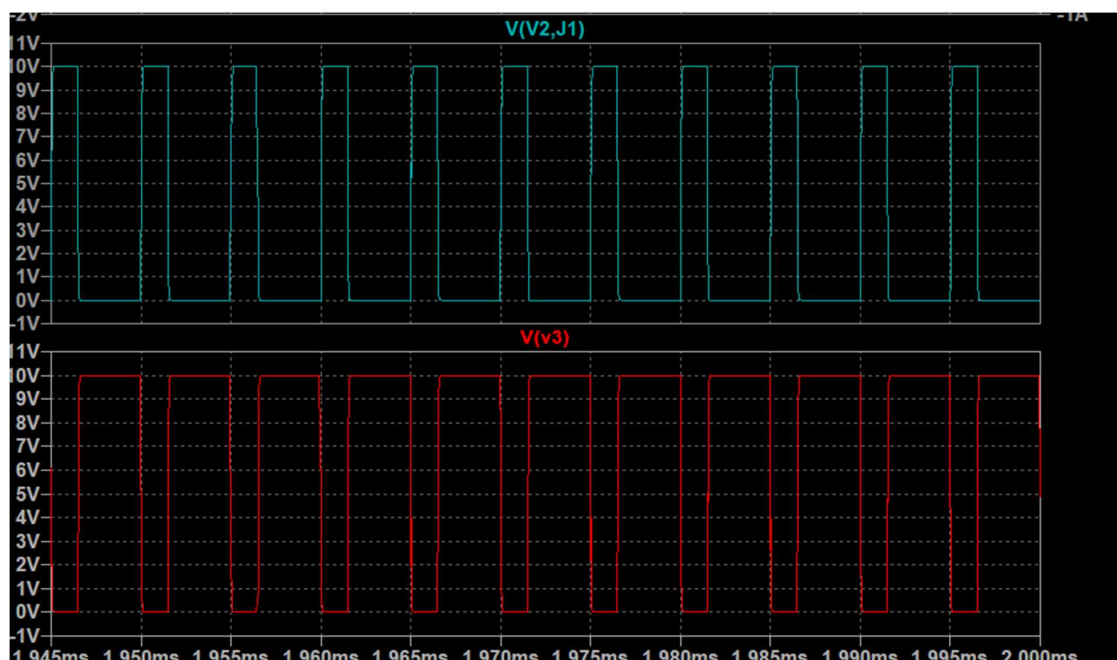
The high-side MOSFET was operated with an ON-time of 1.469 μs , while the low-side MOSFET was configured with a delay time of 1.499 μs and an ON-time of 3.471 μs . During simulation, both MOSFETs were observed to switch in a complementary manner without overlap between their gate pulses.

To verify the dead-time practically, cursor measurements were taken at approximately the 5 V gate-threshold crossing points of the gate-drive waveforms. Although a programmed dead-time of 30 ns was introduced in the pulse generation stage, the effective measured dead-time was found to be approximately 20.4 ns. This difference occurred due to practical switching effects such as finite rise and fall times, gate resistance, and MOSFET parasitic capacitances.

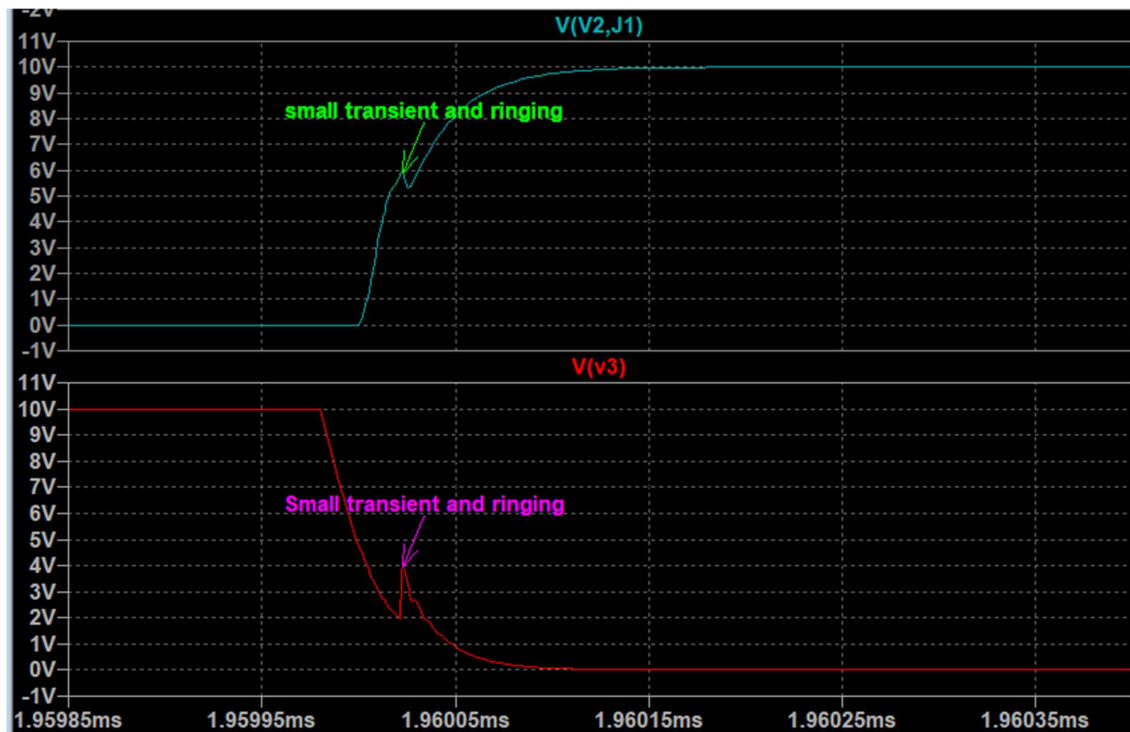
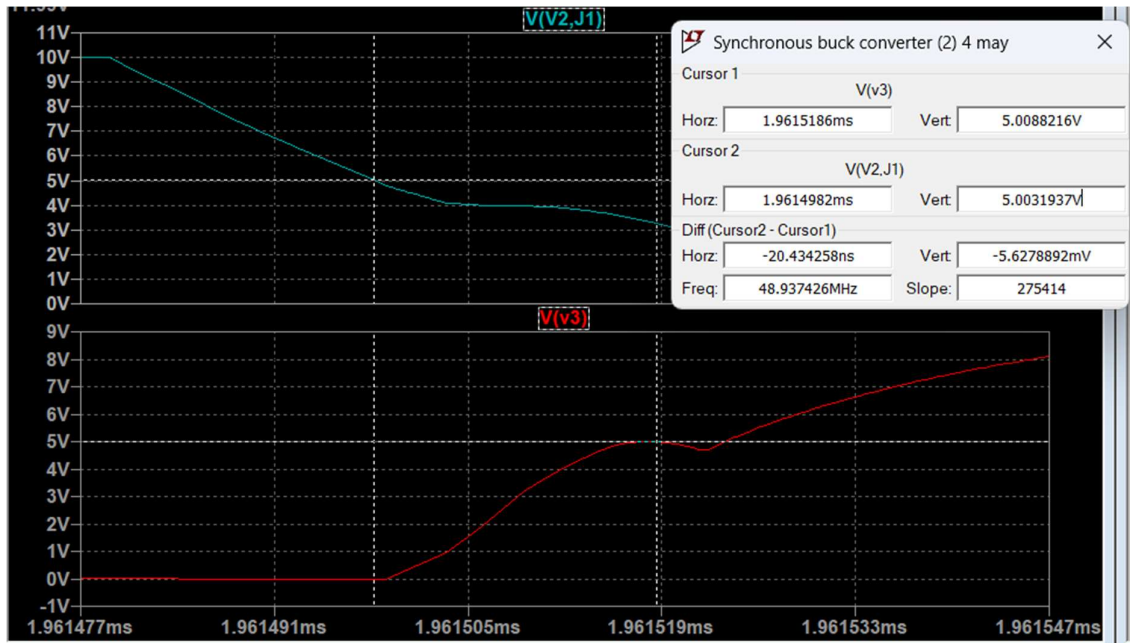
The gate transitions were observed to be smooth rather than ideal instantaneous transitions, indicating realistic switching behaviour in the simulation model. The implemented gate timing successfully ensured stable synchronous switching operation while minimizing shoot-through risk and excessive body-diode conduction.

Observations

- Complementary switching operation was successfully verified.
- No overlap between high-side and low-side gate signals was observed.
- Effective dead-time measured from waveform analysis was approximately 20.4 ns.
- Practical switching effects due to gate resistance and parasitic capacitances were visible in the waveform transitions.
- Stable synchronous switching operation was achieved.



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From the above figure Small transient overshoot and ringing were observed during switching transitions were observed during switching transitions of the gate-drive waveforms. These transients are caused by practical parasitic inductances and MOSFET capacitances interacting during high dv/dt switching events. Reduction of gate resistance improved converter efficiency by reducing switching losses, although it also increased

switching transient magnitude. The RC snubber network helped damp excessive ringing and maintained stable converter operation.

5.2 Switch Node Voltage Analysis: The switch node represents the common connection point between the high-side MOSFET, low-side MOSFET, and output inductor, and therefore experiences high-frequency switching transitions during converter operation.

During simulation, the switch-node voltage was observed to transition between approximately 0 V and the input voltage level in accordance with the complementary switching operation of the MOSFETs. Practical switching effects such as voltage overshoot and ringing were observed during switching transitions due to the presence of parasitic inductance and rapid current variation within the switching loop.

Initially, significant overshoot was observed at the switch node because of input parasitic inductance and fast switching transitions. To reduce this overshoot, switching transient mitigation techniques such as gate resistance optimization and RC snubber implementation were introduced into the converter model.

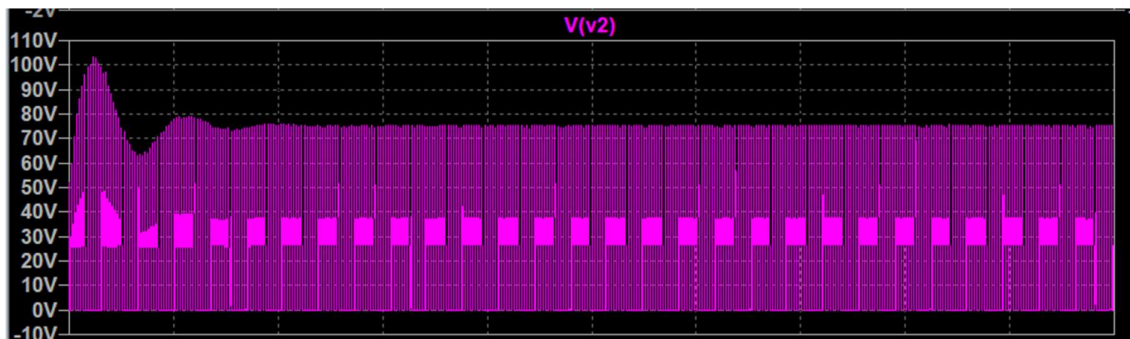
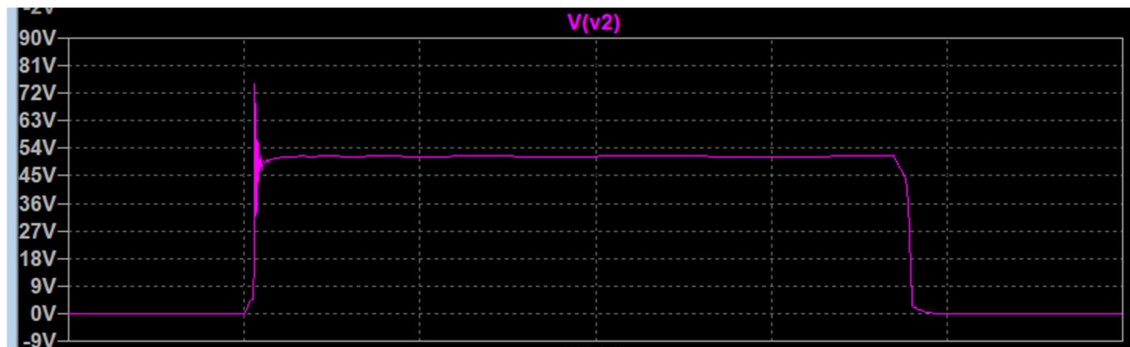
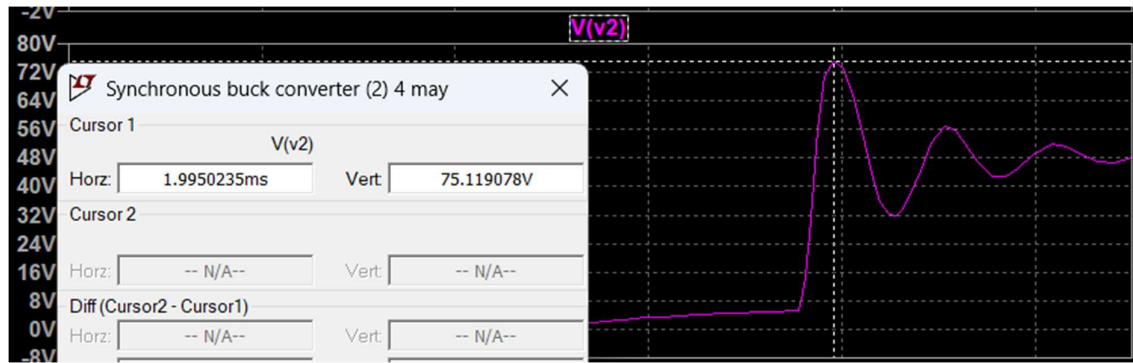
After optimization, the switch-node overshoot was reduced to approximately 75.11 V. The overshoot level remained safely below the drain-to-source voltage rating of the selected MOSFET, indicating acceptable switching stress under the simulated operating conditions. The waveform also showed damped ringing behaviour after each switching transition, which is expected in practical high-frequency switching circuits due to parasitic inductance-capacitance interactions. The implemented snubber network helped reduce the amplitude of this ringing and improved overall switching stability.

The waveform analysis confirmed that the synchronous buck converter achieved stable switching operation while maintaining controlled transient behaviour under practical parasitic conditions.

Observations:

- Switch-node voltage transitioned between approximately 0 V and input voltage level during switching operation.
- Voltage overshoot and ringing were observed due to parasitic inductance and high switching.
- RC snubber network and gate resistance optimization reduced transient overshoot significantly.
- Final measured switch-node overshoot was approximately 75.11 V.
- Stable switching operation was achieved under practical non-ideal conditions.

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A higher switch-node voltage overshoot (Max. 102V < 120V represents approximately 85% utilization of the MOSFET 120V drain to source voltage rating) was observed during startup conditions due to transient charging dynamics, parasitic inductance and high dv/dt switching behavior. The peak startup overshoot remained below the MOSFET drain-to-source voltage rating, while steady state overshoot was significantly lower and well controlled through the RC snubber network and gate resistance optimization.

5.3 Output Voltage analysis: The simulated output voltage varies from 12.07V to 12.02V with an average voltage 12.045V. During steady-state operation, the converter successfully maintained the desired output voltage close to the target value of 12 V. The optimized high-side MOSFET ON-time and synchronous switching operation helped compensate for practical conduction and switching losses present in the converter.

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The output voltage waveform exhibited low ripple characteristics due to the filtering action of the output inductor and capacitor. From waveform measurements, the output voltage varied approximately between 12.07V and 12.02V under steady-state conditions.

The peak-to-peak output voltage ripple was calculated as:

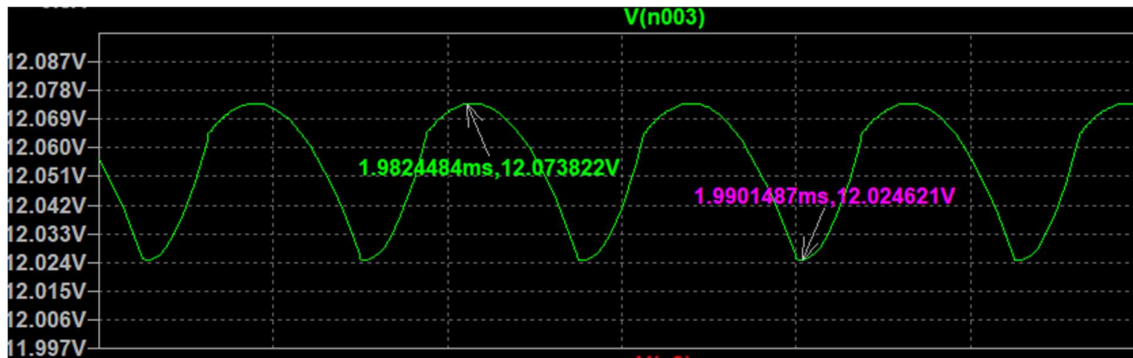
$$V_{\text{ripple}} = 12.07\text{V} - 12.02\text{V} = 50\text{mV}$$

The observed ripple magnitude remained relatively small compared to the average output voltage, indicating effective filtering performance of the LC output stage.

The output waveform remained stable but average steady-state output voltage of 12.07V represents a +0.37% positive deviation from 12V target, which is acceptable for open loop operation. Closed-loop feedback control would eliminate this residual steady-state error

Observations:

- Output voltage was maintained close to the desired 12 V level.
- Stable steady-state operation was achieved.
- Output voltage ripple was approximately 50 mV peak-to-peak.
- LC output filter effectively reduced switching ripple.
- No significant output instability was observed.



5.4 Inductor Current Analysis: Current varies from 4.366A to 5.670A with an average current 5.018A. The inductor current exhibited the expected triangular ripple waveform associated with continuous conduction mode (CCM) operation. During the ON-state of the high-side MOSFET, the inductor current increased as energy was transferred from the input source to the output stage. During the OFF-state of the high-side MOSFET, the low-side synchronous MOSFET provided the freewheeling current path, causing the inductor current to decrease gradually.

From waveform measurements, the inductor current varied approximately between 4.366 A and 5.670 A during steady-state operation.

The peak-to-peak inductor current ripple was calculated as:

$$\Delta I_L = 5.670 - 4.366 = 1.304\text{A}$$

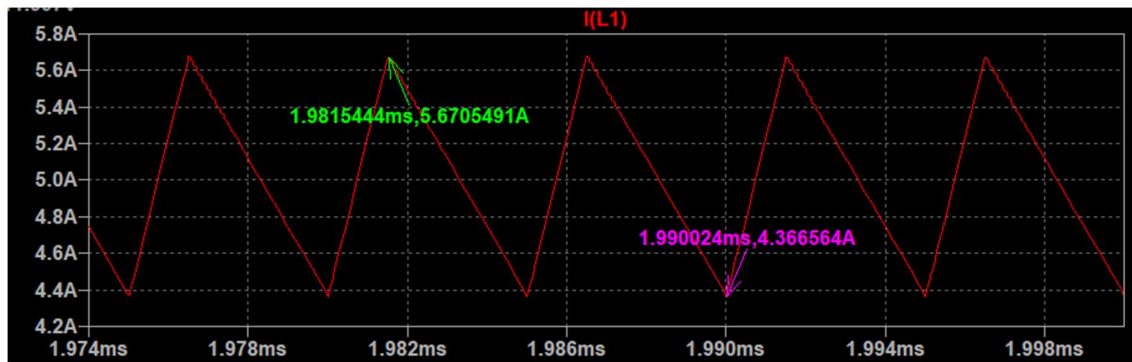
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Since the inductor current never reached zero during the switching cycle, the converter was confirmed to operate in continuous conduction mode. The selected inductance value successfully maintained stable current ripple while supporting the required output load current.

The observed current ripple remained within acceptable limits for the designed operating conditions and contributed to stable converter operation.

Observations

- Inductor current exhibited triangular ripple characteristics.
- Converter operated in continuous conduction mode (CCM).
- Inductor current varied approximately between 4.366 A and 5.689 A.
- Peak-to-peak inductor current ripple was approximately 1.304 A.
- Stable current transfer through synchronous switching operation was achieved.



5.5 Efficiency calculation: The converter efficiency is calculated by comparing average output power to average input power. Output power is determined from the product of average output voltage and load current while input power is determined from the product of average input voltage and source current. The calculated efficiency reflects the cumulative impact of the conduction losses, parasitic resistance, diode forward drop, and switching losses in the practical converter model.

Calculation:

$$\eta(\%) = \frac{P_{out}}{P_{in}} = \frac{V_{out(avg)} * I_{out(avg)}}{V_{in(avg)} * I_{in(avg)}} * 100 = \frac{12.054 * 5.022}{41.846 * 1.534} * 100 = \frac{60.534}{64.192} * 100 = 94.30\%$$

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vin_avg: AVG(V(n001) )=41.8465709623 FROM 0.0015 TO 0.002
iin_avg: AVG(I(V1) )=-1.53428914478 FROM 0.0015 TO 0.002
vout_avg: AVG(V(n003) )=12.0541224398 FROM 0.0015 TO 0.002
iout_avg: AVG(I(R1) )=5.02255101152 FROM 0.0015 TO 0.002
```

- 6. Comparison with Asynchronous Buck Converter:** The performance of the synchronous buck converter was compared with the previously developed asynchronous buck converter to evaluate the effect of synchronous rectification and practical switching optimization on overall converter performance.

In the asynchronous buck converter, a Schottky diode was used as the freewheeling element during the OFF-state of the high-side switch. Although the asynchronous topology offers simpler implementation, the diode forward voltage drop introduces additional conduction losses, especially at higher load currents.

In the synchronous buck converter, the freewheeling diode was replaced with a low-side MOSFET operating in complementary switching mode. This reduced the conduction losses during the freewheeling interval and improved output voltage regulation under practical operating conditions.

The synchronous converter also required proper gate-drive timing optimization and dead-time implementation to prevent shoot-through conditions. Practical parasitic effects such as input loop inductance, MOSFET parasitic capacitances, finite switching transitions, and Miller capacitance effects were observed during simulation and influenced the switching behaviour of the converter.

To reduce switching overshoot and ringing, gate resistance optimization and RC snubber implementation were included in the synchronous converter model. These improvements helped achieve more controlled switching transitions and improved transient performance compared to the asynchronous topology.

A comparative summary of both converter topologies is presented below:

During synchronous converter optimization, it was observed that the ideal theoretical ON-time was insufficient to maintain the desired output voltage under practical non-ideal conditions. Therefore, the high side MOSFET on-time slightly increased to compensate for conduction and switching losses, resulting in improved output voltage regulation.

Sr no.	Parameter	Asynchronous buck converter	Synchronous buck converter	Observation
1	Free wheeling Element	Schottky diode RBQ20NS100A	Low-side MOSFET BSC190N12NS3	Reduced conduction loss
2	Output Voltage	~11.3V	~12.02 – 12.07V	Improved output regulation
3	Switching control	Simpler	Complex	Additional gate timing required
4	Dead-time requirement	Not required	Required	Prevents shoot-through
5	Switching overshoot	Higher	Reduced after optimization	Improved transient control
6	Conduction loss	Higher	Lower	Improved efficiency potential
7	Practical parasitic analysis	Limited	Included	More realistic converter model
8	Switching optimization	Minimal	Gate timing and snubber optimized	Improved switching performance
9	Efficiency	~92.4%	~94.3%	Improved efficiency

7. Loss component analysis:

Loss component	Formula	Estimated value
High Side MOFET Conduction loss	$I_{Avg}^2 * R_{DS(on)} * D$	$5.018^2 * 0.019 * 0.294 = 0.141 \text{ W}$
Low-side MOSFET conduction loss	$I_{Avg}^2 * R_{DS(on)} * (1-D)$	$5.018^2 * 0.019 * 0.706 = 0.338 \text{ W}$
Switching loss	$0.5 * V * I * (t_r + t_f) * f$	$0.5 * 42 * 5.018 * (40.8 \text{ n}) * 200 \text{ k} = 0.86 \text{ W}$
Inductor copper loss	$I_{Avg}^2 * DCR$	$5.018^2 * 0.05 = 1.259 \text{ W}$
Source resistance loss	$I_{In(avg)}^2 * R_s$	$1.53^2 * 0.1 = 0.235 \text{ W}$
Gate drive switching loss	$2 * Q_g * V_{GS} * f_{Sw}$	$2 * 25 \text{ n} * 10 * 200 \text{ k} = 0.10 \text{ W}$
RC Snubber loss	$0.5 * C * V^2 * f$	$0.5 * 2 \text{ n} * 42^2 * 200 \text{ k} = 0.3528 \text{ W}$
Dead-time loss	$V_f * I * t_{dead} * f_{Sw}$	$0.7 * 5.018 * 20.4 \text{ n} * 200 \text{ k} = 0.014 \text{ W}$
Estimated total loss	-	~3.3 W

The analytical loss estimation predicts approx. 94.8% efficiency under simplified assumptions. However, measured LTspice simulation efficiency is comparatively lower due to additional dynamic non-ideal effects including parasitic ringing currents, transient overlap conduction switching waveform non-linearity and practical simulation parasitics not fully captured in analytical calculations.

7.1 Inductor copper loss: A practical inductor DCR of $50\text{m}\Omega$ was assumed to model realistic copper winding losses present in power inductors used in medium-power DC-DC converters. However, this copper loss can be reduced by selecting an inductor with lower DCR and higher current handling capability. Larger wire cross-section and improved magnetic core design help reduce winding resistance and conduction loss.

7.2 Switching loss: Switching losses can be minimized by selecting MOSFETs with lower gate charge, lower output capacitance and faster switching characteristics. Proper gate resistance selection helps optimize switching speed while avoiding excessive ringing and EMI.

7.3 RC Snubber loss optimization: Although the RC snubber reduces switching transients and voltage overshoot, excessive snubber capacitance or low snubber resistance increases power dissipation. Proper optimization of snubber resistor and capacitor values helps achieve a balance between transient suppression and efficiency.

8. Key Learnings

- Practical synchronous buck converters behave differently from ideal theoretical designs due to parasitic elements and switching losses.
- Proper dead-time is important to prevent shoot-through between MOSFETs.
- RC snubber circuits help reduce ringing and voltage spikes but can reduce efficiency if oversized.
- Gate resistance affects switching speed, waveform quality, and switching losses.
- Simulation efficiency may differ from analytical calculations because practical switching effects and parasitics introduce additional losses.
- Synchronous buck converters require optimization between efficiency, stability, and switching performance.
- In this case the inductor copper losses, switching losses and RC snubber loss contributes almost 75% losses which can be reduced by further optimizing by selecting the proper component.

9. Conclusion

A 60W open-loop synchronous buck converter was successfully designed and analyzed in LTspice under practical non-ideal conditions. The converter achieved near 12V output with proper complementary gate-drive operation and dead-time implementation. Practical parasitic elements and snubber circuitry were included to study realistic switching behavior and efficiency tradeoffs. The project improved understanding of synchronous buck converter operation, switching losses, waveform analysis, and practical power electronics design considerations.